

UNITED STATES PATENT AND TRADEMARK OFFICE

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BEFORE THE PATENT TRIAL AND APPEAL BOARD

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GUARDANT HEALTH, INC.,  
Petitioner,

v.

UNIVERSITY OF WASHINGTON,  
Patent Owner.

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IPR2022-01158  
Patent 10,752,951 B2

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Before JOHN G. NEW, ZHENYU YANG, and TINA E. HULSE,  
*Administrative Patent Judges.*

YANG, *Administrative Patent Judge.*

JUDGMENT

Final Written Decision  
Determining All Challenged Claims Unpatentable  
*35 U.S.C. § 318(a)*

Denying in Part and Dismissing in Part Petitioner's Motion to Exclude  
*37 C.F.R. § 42.64(c)*

## I. INTRODUCTION

Guardant Health, Inc. (“Petitioner”) filed a Petition (Paper 3, “Pet.”), seeking an *inter partes* review of claims 1–28 of U.S. Patent No. 10,752,951 B2 (Ex. 1001, “the ’951 patent”). University of Washington (“Patent Owner”) filed a Patent Owner Preliminary Response. Paper 12 (“Prelim. Resp.”). After further briefing (Papers 13, 15), we instituted trial to review the challenged claims (Paper 19, “DI”).

Thereafter, Patent Owner filed a Response to the Petition (Paper 25, “PO Resp.”), Petitioner filed a Reply to Patent Owner’s Response (Paper 33, “Reply”), and Patent Owner filed a Sur-reply to Petitioner’s Reply (Paper 34, “Sur-reply”).

An oral hearing for this proceeding was held on September 13, 2023, and the transcript of that hearing is of record. *See* Paper 46 (“Tr.”).

The Board has jurisdiction under 35 U.S.C. § 6 and issues this Final Written Decision pursuant to 35 U.S.C. § 318 and 37 C.F.R. § 42.73. For the reasons provided below, we find Petitioner has shown, by a preponderance of the evidence, the unpatentability of claims 1–28.

### A. *Real Parties in Interest*

Petitioner identifies itself, Guardant Health, Inc. as the real party in interest. Pet. 9. Patent Owner identifies the University of Washington and TwinStrand Biosciences, Inc. as the real parties in interest. Paper 8, 1.

### B. *Related Matters*

According to the parties, the ’951 patent is asserted against Petitioner in *TwinStrand Biosciences, Inc. et al. v. Guardant Health, Inc.*, 1-21-cv-01126 (D. Del.). Pet. 9; Paper 8, 1.

Petitioner filed a petition in IPR2022-01159, challenging claims 1, 6, 7, and 9–26 of the '951 patent. We denied review in that case.

The parties are also involved in several other *inter partes* review proceedings. *See* Papers 17, 18.

### C. *The '951 Patent and Related Background*

The '951 patent relates to methods of lowering the error rate of massively parallel DNA sequencing using duplex consensus sequencing (“DCS”). Ex. 1001, code (54), 17:17–19.

The '951 patent states massively parallel DNA sequencing “offer[ed] the unique ability to detect minor variants within heterogeneous mixtures.” *Id.* at 1:33–42. It notes the rapid development of clinical applications of deep sequencing in “prenatal screening for fetal aneuploidy, early detection of cancer and monitoring its response to therapy with nucleic acid-based serum biomarkers.” *Id.* at 1:43–50 (internal citations omitted). Deep sequencing, however, had limitations, including “a practical limit of detection . . . imposed by errors introduced during sample preparation and sequencing,” resulting in “approximately 1% of bases [being] incorrectly identified.” *Id.* at 1:61–2:8. The '951 patent states “[t]his background level of artifactual heterogeneity establishes a limit below which the presence of true rare variants is obscured.” *Id.* at 2:9–11.

Acknowledging several attempts aimed at improving the accuracy and sensitivity of sequencing, the '951 patent states that “significant technical artifacts” still result in error rates far higher than the “true mutation frequency.” *Id.* at 2:12–39. According to the '951 patent, “[i]t would be desirable to develop an approach for tag-based error correction.” *Id.* at 2:64–65.

The '951 patent states its DCS approach “greatly reduces errors by independently tagging and sequencing each of the two strands of a DNA duplex.” *Id.*, Abstract. It explains that, because the two strands are complementary, true mutations are found at the same position in both strands. *Id.*

In one embodiment, the DCS method of the '951 patent involves tagging both strands of duplex DNA with a random, yet complementary double-stranded nucleotide sequence, known as a single molecule identifier (“SMI”) sequence. *Id.* at 21:41–45. According to the '951 patent,

The individually tagged strands are then PCR amplified. Every duplicate that arises from a single strand of DNA will have the same SMI, and thus each strand in a DNA duplex pair generates a distinct, yet related population of PCR duplicates after amplification owing to the complementary nature of the SMIs on the two strands of the duplex. Comparing the sequence obtained from each of the two strands comprising a single molecule of duplex DNA facilitates differentiation of sequencing errors from true mutations. When an apparent mutation is, due to a PCR or sequencing error, the substitution will only be seen on a single strand. In contrast, with a true DNA mutation, complementary substitutions will be present on both strands . . . .

Following tagging with a double-stranded SMI and PCR amplification, a family of molecules is obtained that arose from a single DNA molecule; members of the same PCR “family” are then grouped together by virtue of having a common (i.e., the same) SMI tag sequence. The sequences of uniquely tagged PCR duplicates are subsequently compared in order to create a PCR consensus sequence.

*Id.* at 21:51–22:4. The PCR consensus sequences are called single strand consensus sequences (“SSCSs”). *Id.* at 22:11–13.

The '951 patent discloses that SSCSs from two complementary strands of duplex DNA can be identified by virtue of the complementary

SMIs to identify the “partner SMI.” *Id.* at 22:14–17. “Following partnering of two strands by virtue of their complementary SMIs, the sequences of the strands are compared. Sequence reads at a given position are kept only if the read data from each of the two paired strands is in agreement.” *Id.* at 22:21–25.

*D. Illustrative Claim*

Claim 1 is the sole independent claim. It is illustrative of the claimed subject matter and is reproduced below.

1. A method, comprising:

- (a) providing a sample comprising a set of double-stranded polynucleotide molecules, each double-stranded polynucleotide molecule including first and second complementary strands;
- (b) tagging said double-stranded polynucleotide molecules with a set of duplex tags;
- (c) sequencing at least some of said tagged strands to produce a set of sequence reads;
- (d) sorting sequence reads into paired sequence reads and unpaired sequence reads, wherein (i) each paired read corresponds to sequence reads generated from a first tagged strand and a second differently tagged complementary strand derived from an original parent polynucleotide molecule in said set, and (ii) each unpaired read represents a first tagged strand having no second differently tagged complementary strand derived from an original parent polynucleotide molecule represented among said sequence reads in said set of sequence reads; and
- (e) quantifying at least two of (i) said paired sequence reads, (ii) said unpaired sequence reads, (iii) read depth of said paired sequence reads and (iv) read depth of said unpaired sequence reads.

*Id.* at 37:36–59.

*E. Instituted Challenges to Patentability*

We instituted trial to determine whether the challenged claims are unpatentable based on the following grounds:

| Claims Challenged | 35 U.S.C. § <sup>1</sup> | References            |
|-------------------|--------------------------|-----------------------|
| 1–28              | 102                      | Eltoukhy <sup>2</sup> |
| 1–28              | 103                      | Eltoukhy              |

In support of their respective positions, Petitioner relies on the Declaration of John Quackenbush, Ph.D. (Ex. 1002) and Patent Owner relies on the declaration of Rahul Satija, D.Phil. (Ex. 2023).

II. ANALYSIS

*A. Principles of Law*

To prevail in this *inter partes* review, Petitioner “shall have the burden of proving a proposition of unpatentability by a preponderance of the evidence.” 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

Whether the disclosure of a patent satisfies the written description requirement is a question of fact. *Ariad Pharm., Inc. v. Eli Lilly and Co.*, 598 F.3d 1336, 1351 (Fed. Cir. 2010) (en banc). The test for sufficiency of the written description support is “whether the disclosure of the application relied upon reasonably conveys to those skilled in the art that the inventor had possession of the claimed subject matter as of the filing date sought.” *Id.* This “possession” test “requires an objective inquiry into the four corners of

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<sup>1</sup> The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. §§ 102–103, effective March 16, 2013. Because the ’951 patent has an effective filing date after March 16, 2013, the AIA version of §§ 102 and 103 applies.

<sup>2</sup> WO 2015/100427, published July 2, 2015 (Ex. 1040, “Eltoukhy”).

the specification from the perspective of a person of ordinary skill in the art.” *Id.* Possession shown by evidence “outside of the specification is not enough,” and “a description that merely renders the invention obvious does not satisfy the requirement.” *Id.* at 1352. Instead, it is the specification itself that must demonstrate possession. *Id.*

To show anticipation under 35 U.S.C. § 102, each and every claim element, arranged as in the claim, must be disclosed in a single piece of prior art. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). In addition, a reference can anticipate a claim even if it does not expressly spell out all the limitations arranged or combined as in the claim, if a person of skill in the art, reading the reference, would “at once envisage” the claimed arrangement or combination. *Kennametal, Inc. v. Ingersoll Cutting Tool Co.*, 780 F.3d 1376, 1381 (Fed. Cir. 2015).

We analyze the instituted grounds of unpatentability in accordance with these principles.

### *B. Level of Ordinary Skill in the Art*

In determining the level of ordinary skill in the art, various factors may be considered, including the “type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field.” *In re GPAC, Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995). Furthermore, the prior art itself can reflect the appropriate level of ordinary skill in the art. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

Here, Petitioner asserts that an ordinarily skilled artisan “would typically have an advanced degree, such as a Ph.D., with research experience in genomics, molecular biology, bioinformatics, or a related field, or could

have less education but significant professional experience in one or more of these fields.” Pet. 8–9 (citing Ex. 1002 ¶ 38).

Patent Owner states that for the purposes of this proceeding, it does not dispute Petitioner’s definition of the skill level. PO Resp. 2 (citing Ex. 2023 ¶¶ 21, 22).

After reviewing the record, we adopt Petitioner’s definition as it is consistent with the prior art’s demonstration of the level of ordinary skill in the art at the time of the invention.

### *C. Claim Construction*

In an *inter partes* review, we construe a claim term “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b).” 37 C.F.R. § 42.100(b). Under this standard, we construe the claim term “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.*; *see also Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc) (holding that the words of a claim “are generally given their ordinary and customary meaning,” which is “the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention, i.e., as of the effective filing date of the patent application”).

#### 1. “Sequence Reads”

Petitioner argues that, for purposes of this Decision, no claim term needs specific construction beyond its plain and ordinary meaning. Pet. 10–11. Petitioner, however, seeks to provide context for understanding the scope of the term “sequence reads.” Specifically, Petitioner contends that “sequence reads” are produced by sequencing at least some of the tagged



strands, whereas “consensus sequences” are generated by collapsing sequence reads. *Id.* at 11 (citing Ex. 1001, 4:45–47, 20:50–22:25, 23:29–57, 24:46–48, claim 1, claim 6).

In the Decision to Institute, we agreed with Petitioner that “[t]he claims of the ’951 patent as well as the specification make clear that a ‘sequence read’ is different and distinct from a ‘consensus sequence.’” DI 7 (quoting Pet. 11). Patent Owner does not dispute Petitioner’s assertion. PO Resp. 12. Based on the complete record, we see no reason to change our determination of the scope of the term “sequence reads.”

We further resolve the parties’ dispute over the meaning of “raw sequence reads,” a term Patent Owner employs in its papers. Petitioner protests that Patent Owner “interjects the new term ‘raw sequence reads’ into its analysis more than 90 times despite the term never appearing in any of the claims.” Reply 6. Patent Owner explains that it merely uses the term “raw” to describe “sequence reads,” and to distinguish it from “consensus sequences.” Sur-reply 4.

In consideration of Patent Owner’s explanation, we equate the term “raw sequence reads” in Patent Owner’s arguments with the term “sequence reads” in the challenged claims. *See* Ex. 1001, claim 1, step (c) (defining “sequence reads” as generated by “sequencing at least some of said tagged strands”). In other words, the term “raw sequence reads” does not encompass “consensus sequences.” *See* PO Resp. 12 (Patent Owner stating it “does not dispute that a POSA would understand that a ‘consensus sequence’ of the ’951 patent is generated from raw ‘sequence reads’”)

Step (d) of claim 1 recites “sorting sequence reads into paired sequence reads and unpaired sequence reads.” Patent Owner states that it

“does not necessarily agree” with Petitioner that this sorting is limited to sorting raw sequence reads. PO Resp. 12. Patent Owner, however, does not explain its position on what the sorting limitation in step (d) actually encompasses. Given the claim term “sequence reads,” is distinct from, and does not include, “consensus sequences,” we determine step (d) is indeed limited to sorting raw sequence reads.<sup>3</sup>

2. “Paired Sequence Reads” and “Unpaired Sequence Reads”

Step (d) of claim 1 recites

sorting sequence reads into paired sequence reads and unpaired sequence reads, wherein (i) each paired read corresponds to sequence reads generated from a first tagged strand and a second differently tagged complementary strand derived from an original parent polynucleotide molecule in said set, and (ii) each unpaired read represents a first tagged strand having no second differently tagged complementary strand derived from an original parent polynucleotide molecule represented among said sequence reads in said set of sequence reads.

Ex. 1001, 37:45–55.

The terms “differently tagged,” “paired sequence reads,” and “unpaired sequence reads” appear only in the claims, and not in the Specification, of the ’951 patent. Patent Owner argues that “as discussed throughout the ’951 specification, sequence reads are paired by having complementary SMI sequences, *as read off of the sequencer*.” PO Resp. 13 (citing Ex. 1001, 3:52–4:3, 4:12–30, 21:2–5, 26:20–47, Figs. 1, 3(a)–(c); Ex. 2023 ¶ 77); *id.* at 31 (“Sequence reads, therefore, will be paired by the

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<sup>3</sup> In any event, Patent Owner represents that its analysis on written description support is “for sorting paired and unpaired raw sequence reads.” PO Resp. 12.

function of having another sequence read in the sample that has a tag with a complementary sequence.”). Under its proposed claim construction, Patent Owner contends that “reads are not considered unpaired based on whether the complimentary sequence of the original parent polynucleotide is present in the set of sequence reads.” *Id.* at 31–32 (citing Ex. 2023 ¶ 119).

We disagree with Patent Owner’s proposed construction of the terms “paired sequence reads” and “unpaired sequence reads.” Even though the ’951 patent Specification discloses grouping sequence reads “sharing a unique set of SMI tags” into paired families,<sup>4</sup> “[t]he written description part of the specification itself does not delimit the right to exclude. That is the function and purpose of claims.” *Phillips*, 415 F.3d at 1312 (“[T]he claims of a patent define the invention to which the patentee is entitled the right to exclude.”).

Here, claim 1 explicitly defines the terms “paired sequence reads” and “unpaired sequence reads.” Indeed, in the context of paired versus unpaired sequence reads, claim 1 specifies that the complementarity is related to the “original parent polynucleotide molecule,” and not the duplex tags. For example, claim 1 recites that “each unpaired read represents a first tagged strand having *no second differently tagged complementary strand derived from an original parent polynucleotide molecule* represented among said sequence reads in said set of sequence reads.” Ex. 1001, 37:50–55 (emphasis added). Thus, the plain claim language contradicts Patent Owner’s argument

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<sup>4</sup> We note that the ’951 patent Specification discloses “grouping,” but not “sorting,” sequence reads, as recited in claim 1. *See, e.g.*, Ex. 1001, 3:31–33, 4:14–15, 16:47–50. In fact, the term “sorting” is another term that appears only in the claim, and not the Specification, of the ’951 patent.

that “a sequence read will be considered *unpaired based on whether the tag associated with the read does not have a complementary tag* as read by the sequence in the set of sequence reads.” PO Resp. 31 (emphasis added).

In sum, the terms “paired sequence reads” and “unpaired sequence reads” as recited in claim 1 require determining the complementarity of the target sequence strands derived from the “original parent polynucleotide molecule.”

Claim terms need only be construed to the extent necessary to resolve the controversy. *Wellman, Inc. v. Eastman Chem. Co.*, 642 F.3d 1355, 1361 (Fed. Cir. 2011). On this record and for purposes of this Decision, we see no need to address the construction of any other claim term.

#### *D. Analysis*

Petitioner contends that Eltoukhy anticipates and renders obvious claims 1–28 of the ’951 patent. Pet. 35–61. On its face, the ’951 patent claims priority to a series of applications, including application No. 15/660,785 (“the ’785 application”), filed on July 26, 2017, and application No. PCT/US2013/032665 (“the ’665 PCT”), filed on March 15, 2013. Ex. 1001, code (60); *see also* Pet. 14 (listing “’951 Patent Priority Documents”).

Eltoukhy is a PCT application published on July 2, 2015, after the filing date of the ’665 PCT. *See* Ex. 1040, code (43). Nevertheless, Petitioner asserts that Eltoukhy qualifies as prior art because the challenged claims are not entitled to a priority date that predates Eltoukhy’s publication date. Pet. 13. Specifically, Petitioner argues that

the ’785 application does not provide support under section 112 for a method for “(d) sorting sequence reads into paired sequence reads and unpaired sequence reads” and

“(e) quantifying at least two of (i) said paired sequence reads, (ii) said unpaired sequence reads, (iii) read depth of said paired sequence reads and (iv) read depth of said unpaired sequence reads,” recited in claim 1 and required by claims 1–28 of the ’951 patent.

*Id.* at 15–28. Thus, Petitioner concludes that the claims of the ’951 patent are not entitled to an effective filing date that predates Eltoukhy or earlier than July 26, 2017, the filing date of the ’785 application. *Id.* at 34.

Petitioner asserts that Eltoukhy discloses each limitation of each challenged claim. *Id.* at 35–61. In addition, Petitioner argues that the challenged claims would have been obvious over Eltoukhy, because despite “minor differences in verbiage of [the challenged] claims vis-à-vis Eltoukhy, no patentable distinction can be made in view of the Eltoukhy claims and disclosure read as a whole from the perspective of a POSA.” *Id.* at 35.

Patent Owner argues that Eltoukhy does not qualify as prior art because the challenged claims are entitled to at least the filing date of the ’665 PCT, which antedates Eltoukhy. PO Resp. 16–55. Patent Owner further argues that if Eltoukhy is determined to be prior art, Petitioner still fails to show it renders the challenged claims unpatentable. *Id.* at 57–62.

After reviewing the entire record developed at trial, and as explained below, we determine Petitioner has shown, by a preponderance of the evidence, that Eltoukhy qualifies as prior art and that it anticipates the challenged claims.

#### 1. Disclosure of Eltoukhy

Eltoukhy discloses “methods and systems for determining genetic variants (e.g., copy number variation) in a polynucleotide sample.”

Ex. 1040, Abstract. Eltoukhy states that many of the known methods for accurate copy number variation estimation were able to reduce the errors

introduced by the sample preparation and sequencing processes for the molecules that were converted and sequenced, but were “not able to infer the counts of molecules that were converted but not sequenced.” *Id.* ¶¶ 4, 5.

Eltoukhy explains that because “this count of converted b[ut] unsequenced molecules can be highly variable from genomic region to region, these counts can dramatically and adversely affect the sensitivity that can be achieved.” *Id.* ¶ 5.

Eltoukhy discloses a method that tags both halves of the double-stranded DNA. *Id.* ¶ 6 (“If tagged correctly, each original Watson and Crick (i.e., strand) side of the input double-stranded DNA molecule can be differently tagged and identified by the sequencer and subsequent bioinformatics.”). *Id.* According to Eltoukhy, the number of unseen molecules can be estimated based on the number of Pairs (i.e., molecules where both strands were identified) and Singlets (i.e., molecules where only one strand was identified) detected in a particular region. *Id.*

## 2. Effective Filing Date of the Challenged Claims

In an IPR, although Petitioner has the ultimate burden of persuasion to prove unpatentability, Patent Owner must demonstrate entitlement to a priority date when Patent Owner relies on that priority date to overcome an anticipation or obviousness argument. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1379–80 (Fed. Cir. 2015); *see also In re NTP, Inc.*, 654 F.3d 1268, 1276 (Fed. Cir. 2011) (“[A] patent’s claims are not entitled to an earlier priority date because the patentee claims priority. Rather, for a patent’s claims to be entitled to an earlier priority date, the patentee must demonstrate that the claims meet the requirements of 35 U.S.C. § 120.”).

For purposes of this Decision, we focus on step (d) of claim 1, “sorting sequence reads into paired sequence reads and unpaired sequence reads” (“the sorting limitation”). Petitioner argues that the ’785 application (“Schmitt”)<sup>5</sup> lacks the requisite disclosure supporting the sorting limitation. Pet. 17–20. Petitioner points out that nowhere in Schmitt does it recite the term “sorting.” *Id.* at 17. Acknowledging that “adequate support under § 112 does not strictly require *ipsis verbis* description,” Petitioner asserts that Schmitt “simply fails to describe” the sorting limitation. *Id.*

Patent Owner disagrees. PO Resp. 16–33. Relying on Schmitt’s Figure 3(a)–(c), Example 3, Figure 10, and certain other disclosures, Patent Owner argues that Schmitt describes sorting paired raw sequence reads. *Id.* at 17–25. In addition, Patent Owner contends that “by Schmitt’s method of pairing reads, e.g., via Figure 10, Schmitt necessarily sorts reads into paired and unpaired reads because by pairing reads all that is left over is necessarily unpaired reads.” *Id.* at 26 (citing Ex. 2023 ¶ 102). Thus, according to Patent Owner, Schmitt describes sorting unpaired raw sequence reads. *Id.* at 26–33.

After reviewing the entire record developed at trial, and as explained below, even if we were to accept Patent Owner’s argument that Schmitt provides written description support for sorting paired sequence reads, we still find Schmitt does not provide sufficient support for sorting unpaired sequence reads. Thus, Patent Owner has not shown that the challenged

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<sup>5</sup> In their respective analysis, Petitioner cites the ’785 application (Ex. 1031), whereas Patent Owner cites the ’665 PCT (Ex. 1029). It is undisputed that the ’785 application and the ’665 PCT share the same specification. *See* Pet. 34; PO Resp. 1. We refer to both Exhibits 1029 and 1031 as Schmitt.

claims are entitled to an effective filing date that predates the publication date of Eltoukhy.

*a. Paired Sequence Reads*

In support of its argument that Schmitt provides written description support for sorting “paired sequence reads,” Patent Owner relies on Schmitt’s Figure 3(a)–(c), which shows sequence reads “sharing a unique set of SMI tags are grouped into paired families with members having strand identifiers in either the  $\alpha\beta$  or  $\beta\alpha$  orientation. Each family pair reflects one double-stranded DNA fragment.” Ex. 1029 ¶ 14; *see also* PO Resp. 18 (“Schmitt’s Figure 3(a)–(c) depicts grouping multiple raw sequence reads into families and then pairing the reads in the families if the sequence reads have complementary Single Molecule Identifiers (‘SMIs’).”).

Patent Owner also points to Schmitt’s Example 3 as providing written description support for sorting raw sequence reads. *Id.* at 20–21 (citing Ex. 1029 ¶¶ 116, 117). In Example 3, using randomly sheared DNA ends as SMIs, “reads were grouped by SMI with a minimum family size of 1 member.” Ex. 1029 ¶ 116.

Patent Owner further relies on Schmitt’s Figure 10. PO Resp. 24–25. Although the Python code of Figure 10 is for pairing “DCS reads among partner strands” (Ex. 1029, Fig. 10), Patent Owner asserts that it “can pair raw sequence reads” based on complementary SMIs. PO Resp. 24–25 (citing Ex. 1029 ¶¶ 96, 97; Ex. 2023 ¶¶ 95–98).

In addition, Patent Owner argues that Schmitt provides written description support for sorting raw sequence reads because (1) Schmitt describes “pairing raw sequence reads *without* collapsing the raw sequence reads to generate a ‘consensus sequence,’ e.g., an SSCS read,” *see id.*



at 19–20 (citing Ex. 1029 ¶¶ 71, 79, 80); and (2) “Schmitt also repeatedly refers to just ‘reads,’ without reference to whether they are raw sequence reads or consensus sequence reads,” *id.* at 22–24 (citing Ex. 1029 ¶¶ 23, 31, 96, 103, 116, Fig. 10 (Ex. 2010, 1:5–7, 1:30–31, 1:49)).

As Patent Owner recognizes, Schmitt discloses grouping sequences by the SMI tags. *See* PO Resp. 18 (citing Ex. 1029 ¶ 14, Fig. 3(a)–(c)), 20–21 (citing Ex. 1029 ¶¶ 116, 117); *see also* Ex. 1029 ¶ 11 (Schmitt disclosing “grouping the sequenced SMI-target nucleic acid products into families of paired target nucleic acid strands based on a common set of SMI sequences”), ¶ 79 (“[A] set of sequenced SMI-DNA products generated in the methods described above may be grouped into families of paired target nucleic acid strands based on a common set of SMI sequences.”), ¶ 99 (“Following tagging with a double-stranded SMI and PCR amplification, a family of molecules is obtained that arose from a single DNA molecule, members of the same PCR ‘family’ are then grouped together by virtue of having a common (i.e., the same) SMI tag sequence.”).

Claim 1 of the ’951 patent, however, defines each paired sequence read as “correspond[ing] to sequence reads generated from a first tagged strand and a second differently tagged *complementary strand derived from an original parent polynucleotide molecule.*” Ex. 1001, 37:46–50 (emphasis added). Thus, ordinarily, grouping sequence reads<sup>6</sup> by partner SMI tags does not necessarily result in “paired sequence reads,” as defined in claim 1.

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<sup>6</sup> For purposes of this Decision, we assume, without deciding, that, as Patent Owner asserts, Schmitt discloses pairing not only consensus sequences, but also raw sequence reads. *See* PO Resp. 19–24.

Schmitt, however, only considers sequences “when information from both DNA strands was in perfect agreement.” Ex. 1029 ¶ 96; *see also id.* ¶ 105 (“Sequence reads were considered only when the read data from each of the two strands is in perfect agreement.”), ¶ 117 (disclosing “considered only positions at which the sequence from the two strands was in perfect agreement”). Because of this requirement of “perfect agreement,” when Schmitt groups sequence reads based on the complementarity of the SMI tags, the two strands of the target sequence derived from the original parent polynucleotide molecule are also complementary to each other. In other words, in Schmitt, grouping sequence reads by partner SMI tags does in fact result in “paired sequence reads,” as defined in claim 1.

In sum, assuming Schmitt discloses pairing sequence reads, we accept Patent Owner’s argument that Schmitt provides written description support for sorting “paired sequence reads.” It is, however, a different matter for sorting “unpaired sequence reads.” We explain below.

*b. Unpaired Sequence Reads*

Patent Owner asserts that Schmitt provides support for sorting unpaired raw sequence reads because “by Schmitt’s method of pairing reads, e.g., via Figure 10, Schmitt necessarily sorts reads into paired and unpaired reads because by pairing reads all that is left over is necessarily unpaired reads.” PO Resp. 26 (citing Ex. 2023 ¶ 102), 28–29 (discussing Figure 10 and arguing “sequence reads without a matching SMI would be considered unpaired”); *see also id.* at 26 (“[T]here will necessarily be groups comprised of either *paired or unpaired sequence reads*.”). We disagree.

Claim 1 defines the terms “paired sequence reads” and “unpaired sequence reads” based on the complementarity of the target sequence strands

derived from the “original parent polynucleotide molecule.” Ex. 1001, 37:46–55. As explained above, we reject Patent Owner’s contention that a sequence read is considered unpaired solely based on the complementarity of the SMI tags. *See supra* Section II.C.2.

Due to the “errors introduced during sample preparation and sequencing” (Ex. 1029 ¶ 5), there are four outcomes when determining the complementarity of the sequence reads of two strands of SMI-linked target sequence: (1) the first SMI tag and the first strand target sequence are complementary to the second SMI tag and the second strand target sequence, respectively; (2) the first SMI tag is complementary to the second SMI tag but the first strand target sequence is not complementary to the second strand target sequence; (3) the first SMI tag is not complementary to the second SMI tag but the first strand target sequence is complementary to the second strand target sequence; and (4) the first SMI tag is not complementary to the second SMI tag, and the first strand target sequence is not complementary to the second strand target sequence.

It is undisputed that Schmitt groups sequence reads into paired families based on the SMI tags, and Schmitt only considers sequences in which “information from both DNA strands was in perfect agreement.” Ex. 1029 ¶¶ 14, 96; PO Resp. 18. Thus, among the four outcomes, only scenario (1), in which both the SMI tags and the target sequences are complementary, represents a “paired sequence read,” which claim 1 defines as corresponding to “sequence reads generated from a first tagged strand and a second differently tagged complementary strand derived from an original parent polynucleotide molecule.” Ex. 1001, 37:46–50.

Contrary to Patent Owner's argument (*see* PO Resp. 26), in Schmitt, the leftover of "paired sequence reads" is not necessarily "unpaired sequence reads." For example, in scenario (3), the two sequence reads cannot be paired because the SMI tags are not complementary. But, because the two target sequence strands are complementary, they do not represent an "unpaired sequence read," which claim 1 specifically defines as "a first tagged strand having *no second differently tagged complementary strand derived from an original parent polynucleotide molecule.*" Ex. 1001, 37:50–55 (emphasis added). Thus, even if we find Schmitt provides written description support for sorting "paired sequence reads," we reject Patent Owner's argument that "by pairing raw sequence reads, unpaired raw sequence reads are also sorted, making the written description in Schmitt sufficient." *See* PO Resp. 29.

Patent Owner contends "a POSA would understand Schmitt to describe that paired raw sequence reads may not be recovered due to 'inadequate amplification' of one of the two strands of a DNA duplex, resulting in a pool of *unpaired raw sequence reads.*" *Id.* at 27 (citing Ex. 1029 ¶ 105; Ex. 2023 ¶ 104); *see also id.* at 32 ("Schmitt explicitly states that the reason that partner-strand SMIs are not observed is due to "inadequate amplification of the starting DNA library," i.e., failure to amplify one of the two strands of the double-stranded DNA template, *not sequencing errors in the SMI.*"). We are not persuaded by this argument.

Schmitt suggests that "inadequate amplification" is a possible reason for the "low recovery of tag pairs." Ex. 1029 ¶ 105. "Low recovery of tag pairs," however, is not the same as "unpaired sequence reads." We reiterate that, as required in claim 1, whether sequence reads are paired or unpaired is

determined by the complementarity of the target sequence strands derived from the “original parent polynucleotide molecule,” and not the SMI tags. Moreover, even if “inadequate amplification” is a possible reason for “unpaired sequence reads,” it does not detract from the fact that sequencing errors in the SMI leads to “unpaired sequence reads,” as explained below.

Patent Owner emphasizes that “sequencing errors arise at a low rate,” and thus, “any incidence of errors in the SMI would not be the major source of unpaired reads.” *Id.* at 32 (citing Ex. 2032, 1–2; Ex. 2023 ¶ 122); *see also id.* at 27–28 (arguing the frequency of sequencing error is miniscule). We are not persuaded by this argument either.

First, Schmitt discloses that the sequencing errors occur at a rate of “approximately 1%.” Ex. 1029, Abstract; *see also* Ex. 2023 ¶ 121 (Dr. Satija testifying that the barcode sequencing error rate is “between 0.001 to 3%”). That rate, however, as Petitioner points out, is an “average base calling error rate,” and not the frequency of base calling errors in the SMI region. *See* Reply 3 (citing Ex. 1029 ¶ 109).

In Example 1, Schmitt uses a twelve-nucleotide SMI on each end of the target sequence strand. Ex. 1029 ¶ 96. In other words, for each target sequence strand, the SMI tags are twenty-four nucleotides. Dr. Quackenbush testifies that “if the sequencing error rate is only 3 percent, over a 24 nucleotide SMI tag, more than half of them will have sequence-induced errors.” Ex. 2025, 149:6–9; *see also* Reply 3 (“Applying a 1% sequencer error rate results in about 1 in 4 sequence reads with an error in the barcode [of Schmitt’s Example 1].”).

Second, sequencing error rates are not uniformly distributed across sequence reads; instead, most errors occur at the beginning and end of reads.

Ex. 2007, 2. Because the SMI tags are located at the ends of the sequence reads, we find reasonable Petitioner’s argument that Patent Owner’s own evidence indicates “an even higher incidence of SMI sequence errors.”

*See* Reply 3.

Third, Patent Owner’s own evidence teaches that, in addition to sequencing errors, “[e]rrors can also be introduced to sequence tags during tag synthesis and strand replication (library preparation or template amplification).” Ex. 2007, 1; *see also* Ex. 1029 ¶ 14 (Schmitt discussing “PCR-introduced errors”).

Referring to Stoler 2020,<sup>7</sup> Patent Owner contends that actual sequencing data show only 3–5% of the barcodes contained errors. Sur-reply 12 (citing Ex. 1063,<sup>8</sup> 3). Stoler 2020 was published on March 4, 2020 (Ex. 1063, 10), seven years after the priority date Patent Owner contends the challenged claims are entitled to. PO Resp. 16 (arguing “the ’951 patent claims . . . are entitled to at least the filing date of . . . March 15, 2013”); *see also* Sur-reply 10 (Patent Owner faulting Petitioner for “focus[ing] on the *extrinsic, post-filing* Stoler references”).

Written description support “is judged based on the state of the art as of the priority date.” *Biogen Int’l GMBH v. Mylan Pharms. Inc.*, 18 F.4<sup>th</sup> 1333, 1342 (Fed. Cir. 2021). As a result, evidence “illuminating the

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<sup>7</sup> Stoler et al., *Family Reunion Via Error Correction: An Efficient Analysis of Duplex Sequencing Data*, 21:96 BMC BIOINFORMATICS 1–10 (2020) (Ex. 1063).

<sup>8</sup> Patent Owner cites Exhibit 1064, a different Stoler article published in 2016. *See* Sur-reply 12. The context, however, suggests that the correct citation is Exhibit 1063.

state of the art subsequent to the priority date is not relevant to written description.” *Id.* We, thus, do not consider Stoler 2020 in our analysis.<sup>9</sup>

Regardless of the error rate in barcodes, it is undisputed that “the occurrence of sequencing errors in barcodes cannot be completely eliminated.” *See* PO Resp. 32. Acknowledging this, Patent Owner nevertheless argues that “this occurrence can be minimized.” *Id.* (citing Ex. 2023 ¶¶ 122–125). Dr. Satija testifies that “a POSA would have understood that there were many ways to minimize the occurrence of sequencing errors in the SMI.” Ex. 2023 ¶ 122.

We find Dr. Satija’s testimony insufficient to support Patent Owner’s priority claim. The Federal Circuit has instructed that

[e]ntitlement to a filing date does not extend to subject matter which is not disclosed, but would be obvious over what is expressly disclosed. It extends only to that which is disclosed. While the meaning of terms, phrases, or diagrams in a disclosure is to be explained or interpreted from the vantage point of one skilled in the art, all the limitations must appear in the specification. The question is not whether a claimed invention is an obvious variant of that which is disclosed in the specification. Rather, a prior application itself must describe an

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<sup>9</sup> We note that Patent Owner calculates the 3–5% error rate based on Stoler’s finding that “62 and 104 tags out of 2,000 contained errors.” Sur-reply 12 (citing Ex. 1063, 3). The 2,000 tags, however, represent an incomplete set of data. *See* Ex. 1063, 3 (teaching that, for duplex reads, “a total of 2000 tags [were] chosen” after “select[ing] 1000 random combinations” from the “list of 12+12 barcode combinations” in order “to reduce the time required for subsequent computation”). When it “sorted all combinations of barcodes in lexicographic order and counted the number of times each combination appears in this list,” Stoler 2020 found “a striking abundance of combinations that appear only once.” *Id.* Stoler 2020 recognizes that “[s]uch singletons cannot be used in an analysis and are discarded;” yet, “they account for a large fraction of total sequencing output.” *Id.*

invention, and do so in sufficient detail that one skilled in the art can clearly conclude that the inventor invented the claimed invention as of the filing date sought.

*Lockwood v. Am. Airlines, Inc.*, 107 F.3d 1565, 1571–72 (Fed. Cir. 1997); *see also PowerOasis, Inc. v. T-Mobile USA, Inc.*, 522 F.3d 1299, 1310 (Fed. Cir. 2008) (“Obviousness simply is not enough; the subject matter must be disclosed to establish possession.”).

Here, relying on two prior art references, Dr. Satija opines that “a POSA would have known how to apply these techniques by choosing sequences and sequence lengths to minimize sequencing errors from occurring in the SMI.” *Id.* ¶¶ 123–125 (citing Ex. 2007, 1–2; Ex. 2032, 2). Neither Dr. Satija nor Patent Owner, however, points to the ’951 patent Specification as disclosing minimizing sequencing errors in the SMI tags such that scenario (3), that is, the SMI tags are not complementary but the two strands of the target sequence are complementary, does not occur. Thus, even if we accept that only 3–5% of the barcodes contained errors (*see* Sur-reply 12), such numbers undermine Patent Owner’s argument that “by pairing reads all that is left over is necessarily unpaired reads.” *See* PO Resp. 26.

In sum, based on the complete record developed at trial, we find that, even if we accept Patent Owner’s argument that Schmitt provides written description support for sorting “paired sequence reads,” Patent Owner has not shown Schmitt provides written description support for sorting “unpaired sequence reads.”



*c. Summary*

Patent Owner has not shown sufficiently that Schmitt provides written description support for claim 1.<sup>10</sup> Thus, Patent Owner has not demonstrated claim 1 and its dependent claims 2–28 are entitled to the priority of the '665 PCT. As a result, we find Petitioner has met its burden in showing that Eltoukhy qualifies as prior art. *See* Pet. 13–34.

3. Alleged Anticipation by Eltoukhy

Petitioner asserts Eltoukhy anticipates claims 1–28 of the '951 patent. Pet. 35–61. Petitioner provides detailed analysis, mapping each claim limitation to the disclosures of Eltoukhy. *Id.* After reviewing the entire record developed at trial, and as explained below, we determine that Petitioner has shown, by a preponderance of the evidence, that Eltoukhy discloses each limitation of the challenged claims as arranged therein.

*a. Claim 1*

We focus our analysis on claim 1, the only independent claim of the '951 patent.

(1) Steps (a)–(d)

It is undisputed that steps (a), (c), and (d) of challenged claim 1 are identical (or substantively identical) to steps (a), (c), and (e), respectively, of Eltoukhy's claim 87. *Compare* Ex. 1001, claim 1, *with* Ex. 1040, claim 87.

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<sup>10</sup> The parties also dispute whether Schmitt provides written description support for the quantifying step (e) of claim 1 as well as numerous dependent claims. Pet. 20–34; PO Resp. 33–52. Because we find Patent Owner has not shown written description support for the sorting step (d) of claim 1, we do not need to address the additional arguments.

Step (b) of challenged claim 1 recites “tagging said double-stranded polynucleotide molecules with a set of duplex tags.” Ex. 1001, 37:41–42. In addition to this same language, step (b) of Eltoukhy’s claim 87 further recites “wherein each duplex tag differently tags said first and second complementary strands of a double-stranded polynucleotide molecule in said set.” Ex. 1040, claim 87. Petitioner contends that this additional limitation recited in Eltoukhy is within the scope of step (b) of challenged claim 1. Pet. 36–37 (citing Ex. 1002 ¶ 108). We agree with Petitioner, and Patent Owner does not dispute this assertion.

(2) Step (e)

Step (e) of challenged claim 1 recites “quantifying at least two of (i) said paired sequence reads, (ii) said unpaired sequence reads, (iii) read depth of said paired sequence reads and (iv) read depth of said unpaired sequence reads.” Ex. 1001, 37:56–59. Step (f) of Eltoukhy’s claim 87 recites “determining quantitative measures of at least two of (i) said paired reads, (ii) said unpaired reads that map to each of one or more genetic loci, (iii) read depth of said paired reads and (iv) read depth of unpaired reads.” Ex. 1040, claim 87.

Petitioner observes that step (f)(ii) of Eltoukhy’s claim 87 recites “that map to each of one or more genetic loci,” whereas the corresponding step (e)(ii) of challenged claim 1 does not. Pet. 39. According to Petitioner, this difference renders challenged claim 1 narrower than Eltoukhy’s claim 87. *Id.* (additionally arguing that the recitation of this phrase in some dependent claims “indicating that Patent Owner intended claim 1 to encompass the scope of claim 87 of Eltoukhy”). We agree with Petitioner, and Patent Owner does not dispute this assertion.

Patent Owner, however, emphasizes that challenged claim 1 uses the word “quantifying,” whereas Eltoukhy’s claim 87 uses the phrase “determining quantitative measures.” PO Resp. 59–62. Patent Owner argues that, based on the plain and ordinary meaning of the term “quantifying,” “a POSA would understand the ‘quantifying’ limitation in steps (e)(i) and (e)(ii) of claim 1 to require counting the number of paired and unpaired raw sequence reads.” *Id.* at 14 (citing Ex. 2023 ¶ 82). For purposes of this Decision, we accept, without deciding, Patent Owner’s proposed construction of “quantifying” paired and unpaired sequence reads.

Patent Owner also contends that Eltoukhy discloses “quantitative measures” as a genus of “different measures and not specifically counting the number of . . . paired and unpaired reads.” *Id.* at 60 (citing Ex. 1040 ¶¶ 35, 252, 258; Ex. 2023 ¶¶ 204–210). For purposes of this Decision, we accept, without deciding, Patent Owner’s contention on this issue, too.

But even after accepting Patent Owner’s two assertions, we still find Eltoukhy anticipates claim 1. It is correct that a genus does not always anticipate a claim to a species within the genus; but when the species is clearly named, the species claim is anticipated no matter how many other species are additionally named. *See In re Gleave*, 560 F.3d 1331, 1338 (Fed. Cir. 2009).

In this case, as Petitioner correctly points out, “Eltoukhy expressly describes counting as a quantitative measure.” Reply 23–24 (citing Ex. 1040 ¶¶ 183, 184, 186). Indeed, Eltoukhy discloses its method comprises “quantifying sequence reads” and states quantifying includes counting or estimating the number. *See* Ex. 1040 ¶ 183 (“Pairs and Singlets mapping to a selected genetic locus, or to each of a plurality of selected genetic loci, are

quantified, *e.g.*, counted.”), ¶ 184 (“The quantifying can comprise estimating number of polynucleotides in the sample [*e.g.*, Pairs polynucleotides, Singlets polynucleotides, or Unseen polynucleotides.”), ¶ 186 (“The number of Unseen original molecules in a sample can be estimated based on the relative number of Pairs and Singlets.”).

Patent Owner does not appear to dispute that Eltoukhy discloses counting as a quantitative measure. *See* Sur-reply 7 (“PO did not state that Eltoukhy does not disclose ‘counting.’”), 25 (the same). Instead, Patent Owner insists that Petitioner “has not met its burden to show that [Petitioner] Guardant’s reliance on Eltoukhy’s broad genus of ‘determining quantitative measures,’ without more, is sufficient to anticipate . . . the claimed ‘quantifying.’” *Id.* at 7–8; *see also id.* at 25 (contending that Petitioner has failed to “address how the broad genus of ‘quantitative measures’ disclosed in Eltoukhy is sufficient to anticipate the specific quantifying recited in claim 1”); PO Resp. 61 (the same).

We disagree with Patent Owner’s characterization of Petitioner’s arguments. Petitioner’s mapping of the claimed “quantifying” to Eltoukhy does not solely rely on a genus “without more.” Rather, Petitioner cites to paragraphs where Eltoukhy specifically discloses counting as quantifying. *See, e.g.*, Reply 23–24 (citing Ex. 1040 ¶¶ 183, 184, 186).

During oral argument, counsel for Patent Owner complained that Petitioner did not cite paragraph 183 of Eltoukhy until the Reply stage. Tr. 46:1–8. We are not persuaded by this argument. First, the Petition cites paragraph 183 of Eltoukhy. *See* Pet. 4 (“Eltoukhy describes methods that combine sample processing and bioinformatic techniques to more accurately

detect and analyze sample molecules, including unseen molecules, in copy number variation analysis. EX1040, Figures 1–2, ¶¶ 6, 183–187.”).

Second, in the claim chart mapping the “quantifying” step, the Petition cites Figure 1 of Eltoukhy. *Id.* at 38–39 (citing Ex. 1040, Fig. 1). Step 114 in Figure 1 describes “Determine quantitative measure of Pairs and Singlets mapping to each selected locus.” Ex. 1040, Fig. 1. Paragraph 183 merely explains that, in this specific step, quantifying is counting. *See* Ex. 1040, ¶ 183 (“For example, as shown in step (114) (FIG. 1), Pairs and Singlets mapping to a selected genetic locus, or to each of a plurality of selected genetic loci, are quantified, *e.g.*, counted.”).

Third, even if the Petition did not explicitly cite paragraph 183 of Eltoukhy, Petitioner may properly do so at the Reply stage to respond to Patent Owner’s newly proposed construction, limiting “quantifying” to “counting the number.” *Parkervision, Inc. v. Vidal*, 88 F.4th 969, 979–80 (Fed. Cir. 2023); *Axonics, Inc. v. Medtronic, Inc.*, 75 F.4th 1374, 1383–84 (Fed. Cir. 2023).

In *Axonics*, neither the petitioner nor the patent owner expressly proposed a pre-institution construction of any claim terms, and the Board’s institution decision did not expressly construe any terms. 75 F.4th at 1378. Post-institution, the patent owner response proposed a construction of a claim limitation, which the final written decision ultimately adopted. *Id.* at 1378–79. The petitioner’s reply contended that its relied-upon prior art taught the disputed limitation, even under the patent owner’s newly-proposed construction, but the Board refused to consider that argument because the petition had not included such an argument. *Id.* at 1379. The Federal Circuit vacated the Board’s decision, holding that

“where a patent owner in an IPR first proposes a claim construction in a patent owner response, a petitioner must be given the opportunity in its reply to argue and present evidence of anticipation or obviousness under the new construction, at least where it relies on the same embodiments for each invalidity ground as were relied on in the petition.” *Id.* at 1384.

Similarly, in *Parkervision*, the Federal Circuit explained that “[b]ecause the Board’s institution decision did not contain an express claim construction, the APA required the Board to give [the petitioner] Intel ‘adequate notice and an opportunity to respond under the new construction’ first proposed in ParkerVision’s patent owner response.” 88 F.4th at 979.

Here, as in *Parkervision* and *Axonics*, neither party expressly proposed a pre-institution construction of the term “quantifying,” and we did not construe the term in the Decision to Institute either. Post-institution, Patent Owner proposes that we limit “quantifying” in steps (e)(i) and (e)(ii) of claim 1 to require counting the number of paired and unpaired raw sequence reads (PO Resp. 14), which we accept for purposes of this Decision. In the Reply, Petitioner contends that Eltoukhy discloses “quantifying,” even under Patent Owner’s newly proposed construction of the term. Reply 8, 23–24.

As explained above, paragraph 183 elucidates Figure 1. Ex. 1040 ¶ 183. Similarly, paragraphs 184 and 186 also explain Figures 1 and 2, respectively. *Id.* ¶¶ 184, 186. The Petition cites Figures 1 and 2 of Eltoukhy. Pet. 38–39 (citing Ex. 1040, Figs. 1, 2). Under *Parkervision* and *Axonics*, Petitioner’s argument and reliance on paragraphs 183, 184, and 186, which address Figures 1 and 2, are proper. *See Parkervision*, 88 F.4th at 979 (holding the petitioner’s reply “appropriately responded” to the patent

owner's new claim construction by explaining how the prior art discloses a claim limitation even under that proposed construction); *Axonics*, 75 F.4th at 1384 (holding the reply is proper because the petitioner "in the reply, relied on the same embodiments as it relied on in the petition").

Thus, even if we limit "quantifying" in steps (e)(i) and (e)(ii) of claim 1 to "counting the number," and even if we accept Patent Owner's argument that Eltoukhy discloses "quantitative measures" as a genus of "different measures," Petitioner has shown that Eltoukhy specifically discloses counting the numbers of paired and unpaired sequence reads. *See* Pet. 38–39 (citing Ex. 1040, Figs. 1, 2, ¶¶ 5, 6, 54, 163–165); Reply 23–24 (citing Ex. 1040 ¶¶ 183, 184, 186).

Alternatively, we note whether a generic disclosure necessarily anticipates everything within the genus "depends on the factual aspects of the specific disclosure and the particular products at issue." *Sanofi-Synthelabo v. Apotex, Inc.*, 550 F.3d 1075, 1083 (Fed. Cir. 2008). A prior art reference may be deemed to disclose each member of a genus when, reading the reference, an ordinarily skilled artisan can "at once envisage each member of this limited class." *In re Petering*, 301 F.2d 676, 681 (CCPA 1962). In this context, "how one of ordinary skill in the art would understand the relative size of a genus or species in a particular technology is of critical importance." *Osram Sylvania Inc. v. American Induction Tech. Inc.*, 701 F.3d 698, 706 (Fed. Cir. 2012).

Patent Owner repeatedly asserts that the "quantitative measures" disclosed in Eltoukhy is a "broad" genus. *See* PO Resp. 60–61; Sur-reply 2, 7, 25. As support, Patent Owner cites the Declaration of Dr. Satija. PO Resp. 60–61 (citing Ex. 2023 ¶ 208). Dr. Satija testifies that

quantitative measures for tagged polynucleotides known in the art [at relevant time] would have included a broad genus of measurements, including, but not limited to, quantifying mutation rate, read length, G-C and A-T polynucleotide content, the number of paired end reads with just a singleton read, number of read pairs with reads from both ends present, allele frequency, read distribution across various chromosomes, etc.

Ex. 2023 ¶ 208.

Dr. Satija, however, does not cite any support for his opinion. Moreover, Dr. Satija lists only a few of the alleged species of the genus. Filler language such as “including, but not limited to” and “etc.” does not help us evaluate the relative size of the genus in order to determine whether an ordinarily skilled artisan can “at once envisage” counting as a species in the genus of “quantitative measures.” As a result, Dr. Satija’s testimony on this issue has limited value to support Patent Owner’s argument.

In sum, Petitioner has shown, by a preponderance of the evidence, that Eltoukhy discloses step (e) of claim 1.

(3) Written Description Support and Enablement of Eltoukhy

Patent Owner also contends that

to the extent that the Board determines that Schmitt does not provide written description support because reads that cannot be paired due to sequencing errors in the barcode should not be considered “unpaired reads,” then Eltoukhy cannot anticipate the ’951 patent claims. This is because Eltoukhy, like Schmitt, relies on barcodes to sort paired and unpaired reads . . . As such, Eltoukhy would suffer from the same alleged defect as Schmitt. *In re Donohue*, 766 F.2d 531, 533 (Fed. Cir. 1985) (“[E]ven if the claimed invention is disclosed in a printed publication, that disclosure will not suffice as prior art if it is not enabling.”).

PO Resp. 58.



It is unclear whether Patent Owner challenges the written description support or the enablement requirement of *Eltoukhy*. If the former, Patent Owner's argument is inapposite. A prior art reference can anticipate a claim despite the fact that the description provided in the anticipating reference might not otherwise entitle its author to a patent. *See Vas-Cath Inc. v. Mahurkar*, 935 F.2d 1555, 1562 (Fed. Cir. 1991) (discussing the “distinction between a written description adequate to *support* a claim under § 112 and a written description sufficient to *anticipate* its subject matter under § 102(b)”).

Here, we do not need to resolve whether, as Patent Owner asserts, “*Eltoukhy*, like *Schmitt*, relies on barcodes to sort paired and unpaired reads.” *See* PO Resp. 58. Even if that is the case, as explained above, *Eltoukhy* still discloses all the limitations of challenged claim 1, mostly with language identical (or substantively identical) to that of claim 1. *See supra* Sections II.D.3.a.(1)–(2). And that is sufficient for anticipation purposes. *See In re Lukach*, 442 F.2d 967 (CCPA 1971) (holding that a U.S. grandparent application did not sufficiently describe the later-claimed invention, but that the appellant's intervening British application, a counterpart to the U.S. application, anticipated the claimed subject matter).

If Patent Owner instead is challenging *Eltoukhy* as not enabled, we are not persuaded. Indeed, there is a presumption of enablement for prior art. *Amgen Inc. v. Hoechst Marion Roussel, Inc.*, 314 F.3d 1313, 1354–55 (Fed. Cir. 2003) (holding a patent challenger is entitled to rely on the presumption of enablement of a prior art patent, and patent owner bears the burden of proving nonenablement); *In Re Antor Media Corp.*, 689 F.3d 1282, 1289 (Fed. Cir. 2012) (holding the same presumption and burden apply to prior art

publication). Here, Patent Owner has not presented any evidence or argument beyond its conclusory assertion that Eltoukhy is not enabled. *See* PO Resp. 58.

(4) Summary

After reviewing the record, we determine Petitioner has shown that Eltoukhy discloses each limitation of claim 1 as arranged therein. Thus, for the reasons discussed above, we determine Petitioner demonstrates by a preponderance of the evidence that Eltoukhy anticipates claim 1.

*b. Claims 2–28*

Petitioner provides further analysis, mapping the additional limitations of dependent claims 2–28 to the disclosures of Eltoukhy. Pet. 40–61. Patent Owner points out that (1) claims 5–7, 11–13, and 15–26 do not narrow the quantification methods recited in step (e) of claim 1; and (2) each of claims 2–4, 8–10, 14, 27, and 28 specifically recites quantifying paired and unpaired sequence reads. PO Resp. 62. Relying on the same argument it advances in addressing the “quantifying” step (e) of claim 1, Patent Owner contends that Petitioner has failed in showing Eltoukhy anticipates claims 2–28. *Id.* For the same reasons as explained above, we reject Patent Owner’s argument. *See supra* Section II.D.3.a.(2).

After reviewing the record, we find Petitioner’s analysis persuasive and adopt it as our own. *See* Pet. 40–61. Thus, we determine Petitioner demonstrates by a preponderance of the evidence that Eltoukhy anticipates claims 2–28.<sup>11</sup>

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<sup>11</sup> Petitioner also argues that the challenged claims would have been obvious over Eltoukhy. Pet. 35. Having determined that Petitioner has shown Eltoukhy anticipates all those claims, we do not address the obviousness

### III. PETITIONER’S MOTION TO EXCLUDE

Petitioner filed a Motion to Exclude Exhibits 2001, 2002, 2004–2006, 2008, 2009, 2011–2020, and 2023. Paper 38 (“MTE”), 1. Patent Owner filed an Opposition to the MTE (Paper 40, “MTE Opp.”), and Petitioner filed a Reply in support of the MTE (Paper 41).

Petitioner, as the party moving to exclude evidence, bears the burden of proving that it is entitled to the relief requested, namely, that the material sought to be excluded is inadmissible under the Federal Rules of Evidence. *See* 37 C.F.R. §§ 42.20(c), 42.62(a); *see also* Consolidated Trial Practice Guide<sup>12</sup> (“CTPG”) 8, 37 (explaining that a motion to exclude is based on inadmissibility).

For the reasons explained below, we dismiss in part and deny in part Petitioner’s MTE.

#### A. *Exhibit 2001*

Exhibit 2001 is the Data Sheet for TwinStrand Duplex Seq AML-29 MRD v1.0 Assay. Patent Owner filed Exhibit 2001 with the Preliminary Response to support its argument relating to the construction of the claim

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issue. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding a petitioner “is entitled to a final written decision addressing all of the claims it has challenged”); *Bos. Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020) (recognizing that the “Board need not address issues that are not necessary to the resolution of the proceeding,” and thus, agreeing that the Board has “discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims”).

<sup>12</sup> Available at <https://www.uspto.gov/TrialPracticeGuideConsolidated>.

term “double-stranded circulating nucleic acid molecules.” Prelim. Resp. 4–5 (citing Ex. 2001, 3).

In a concurrently filed petition challenging claims 1, 6, 7, and 9–26 of the ’951 patent, Petitioner represented that it “does not oppose Patent Owner’s interpretation of the term for prior art unpatentability analysis in these IPRs against the ’951 patent and its related patents.”<sup>13</sup> IPR2022-01159, Paper 3, 17; *see also* Pet. 13 (arguing the priority documents do not provide written description support “[e]ven if [Patent Owner’s] expansive claim construction is adopted”).

Because the issue for which Exhibit 2001 was submitted is not in dispute in this case, we do not rely on this Exhibit in rendering this Decision. Therefore, we dismiss the MTE in this regard.

*B. Exhibit 2002*

Exhibit 2002 is the declaration of Joshua P. Earl, Ph.D., filed in support of Patent Owner’s Preliminary Response. On April 11, 2023, Patent Owner affirmatively withdrew Dr. Earl’s testimony. *See* Ex. 3001. Because we do not rely on Exhibit 2002 in rendering this Decision, we dismiss the MTE in this regard.

*C. Exhibit 2023*

Exhibit 2023 is the Declaration of Rahul Satija, D.Phil. Petitioner moves to exclude Dr. Satija’s proffered opinions because they “are not ‘the product of reliable principles’ and methods ‘reliably applied’ to the facts of

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<sup>13</sup> Patent Owner also points out that in the pending district court case, the parties agreed on the construction of the term “double-stranded circulating nucleic acid molecules.” PO Resp. 15–16 (citing Ex. 2022, 2); MTE Opp. 5.

this case.” MTE 5. According to Petitioner, “Dr. Satija’s declaration testimony is predicated on erroneous legal standards for written description support in the ’951 patent priority documents and regarding certain claim terms.” *Id.* We are not persuaded.

“A motion to exclude must explain why the evidence is not admissible (e.g., relevance or hearsay).” *See* CTPG 79. Petitioner’s arguments in the MTE, however, do not address the admissibility of the Satija Declaration. Instead, they amount to challenging the sufficiency of, or the weight to be given to, the evidence. A motion to exclude is not a proper vehicle for these purposes. *See id.* Therefore, we deny the MTE with regard to Exhibit 2023.

*D. Exhibits 2004–2006, 2008, 2009, and 2011–2020*

Exhibits 2004–2006, 2008, and 2009 are prior art references. Exhibits 2011–2013 and 2017–2020 are documents filed in other IPR proceedings and district court cases. Exhibits 2014–2016 are file histories of applications in the family of the challenged ’951 patent. In rendering this Decision, we do not rely on these Exhibits.

According to Patent Owner, however, Dr. Satija considered Exhibits 2004–2006, 2008, 2009, and 2011–2020 when forming the basis of his expert testimony. MTE Opp. 2–3 (citing Ex. 2023 ¶¶ 20, 107, 108). Because we address the Declaration of Dr. Satija (Exhibit 2023), for the sake of completeness, we deny the MTE with regard to Exhibits 2004–2006, 2008, 2009, and 2011–2020.

#### IV. CONCLUSION<sup>14</sup>

After reviewing the entire record and weighing evidence offered by both parties, we determine that Petitioner has met its burden to show, by a preponderance of the evidence, that Eltoukhy anticipates claims 1–28.

In summary:

| <b>Claims</b>              | <b>35<br/>U.S.C.<br/>§</b> | <b>References</b> | <b>Claims<br/>Shown<br/>Unpatentable</b> | <b>Claim(s) Not<br/>Shown<br/>Unpatentable</b> |
|----------------------------|----------------------------|-------------------|------------------------------------------|------------------------------------------------|
| 1–28                       | 102                        | Eltoukhy          | 1–28                                     |                                                |
| 1–28                       | 103 <sup>15</sup>          | Eltoukhy          |                                          |                                                |
| <b>Overall<br/>Outcome</b> |                            |                   | 1–28                                     |                                                |

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<sup>14</sup> Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. §§ 42.8(a)(3), (b)(2).

<sup>15</sup> As explained above, because we determine that Petitioner has shown Eltoukhy anticipates claims 1–28, we do not address whether these claims would have been obvious over Eltoukhy. *See supra* Section II.D.3.b. n.10.

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has shown, by a preponderance of the evidence, that claims 1–28 of the '951 patent are unpatentable;

FURTHER ORDERED that Petitioner's Motion to Exclude is *denied-in-part* and *dismissed-in-part*;

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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